Design, Configuration, and Simulation of a Hybrid Wired–Wireless Network Using Cisco Packet Tracer

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1. Introduction

In this report, the design, architecture, and simulation of a hybrid computer network based on wired and wireless communication systems are given. It was done in Cisco Packet Tracer as per the CN7029 coursework brief and it was saved in the form of a file named Network_u2848393.pkt. The network will include two Cisco 1941 routers connected by a serial point to point link, 2960 switches serving the wired segments, an AP-PT access point to offer a wireless subnet, three wireless laptops, two servers and seven wired hosts spread across two wired subnets (Basar *et al.*, 2023). The major tasks in this work were the calculation and allocation of IP addresses based on a block exclusively assigned to the students, the design of subnetworks according to the needs of hosts, the application of wireless security WPA2, configuration of DHCP servers to provide wireless customers, route between subnets, and checking complete network connectivity with the help of debugging tools, ping, and PDU simulation (Mwansa *et al.*, 2024).

It is imperative that before laying out the design, we give a background on the principles of wireless which in particular is needed by the assignment. The human readable name that is being broadcast by a wireless access point is known as the Service Set Identifier (SSID) which is the name that is used to identify and connect to a particular network by the wireless devices used. It is similar to a network name tag which the client picks when they scan their Wi-Fi options. WPA2 (Wi-Fi Protected Access 2)- This is the standard encryption and authentication system in the wireless networks of IEEE 802.11. WPA2-PSK involves the use of Advanced Encryption Standard (AES) and a shared key in order to make sure that only the authorised users can join a network, and provide confidentiality of traffic carried via the air (Barnes, 2022). Wireless security is important as radio communication does not remain confined within the physical boundaries of a building, without encryption attackers can intercept information or even gain access without authorization. To fulfill these critical requirements, AP-PT was given SSID UNI_WLAN_A and the pass phrase was P@ssword123 as the secure password using the WPA2-PSK. The rest of the report describes how the computation of block IP address, subnet calculation, network design, device configuration, a simulation, and general evaluation of the hybrid topology implemented (Kanyimama and Argungu, 2023).

2. Student Block IP Address Calculation

2.1 Deriving Unique IP block

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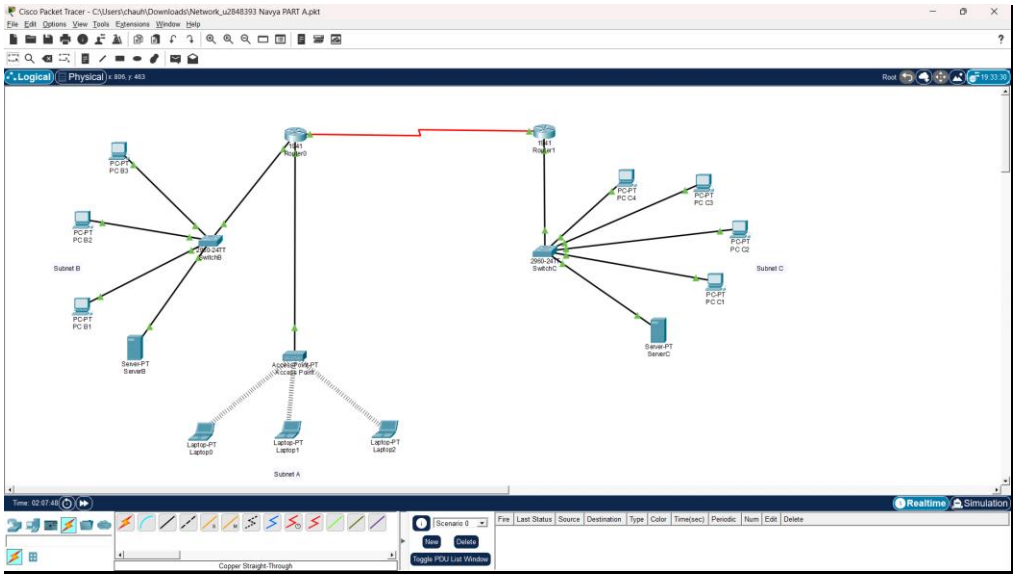
Name	Network	Mask	Usable Hosts
Subnet A (Wireless)	**191.83.93.0/29**	255.255.255.248	.1–.6 (gateway .1)
Subnet B (Wired)	**191.83.93.8/29**	255.255.255.248	.9–.14 (gateway .9)
Subnet C (Wired)	**191.83.93.16/29**	255.255.255.248	.17–.22 (gateway .17)
Serial link (R0–R1)	**191.83.93.24/29**	255.255.255.248	.25–.30


Cisco Packet Tracer - C:\Users\chauh\Downloads\Network_u2848393 Navya PART A.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
Root
PC-PT
PC B3
PC-PT
PC B2
PC-PT
PC B1
Subnet B
Server-PT
ServerB
PC-PT
PC C4
PC-PT
PC C3
PC-PT
PC C2
PC-PT
PC C1
Subnet C
Server-PT
ServerC
Laptop-PT
Laptop0
Laptop-PT
Laptop1
Laptop-PT
Laptop2
Subnet A
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Copper Straight-Through

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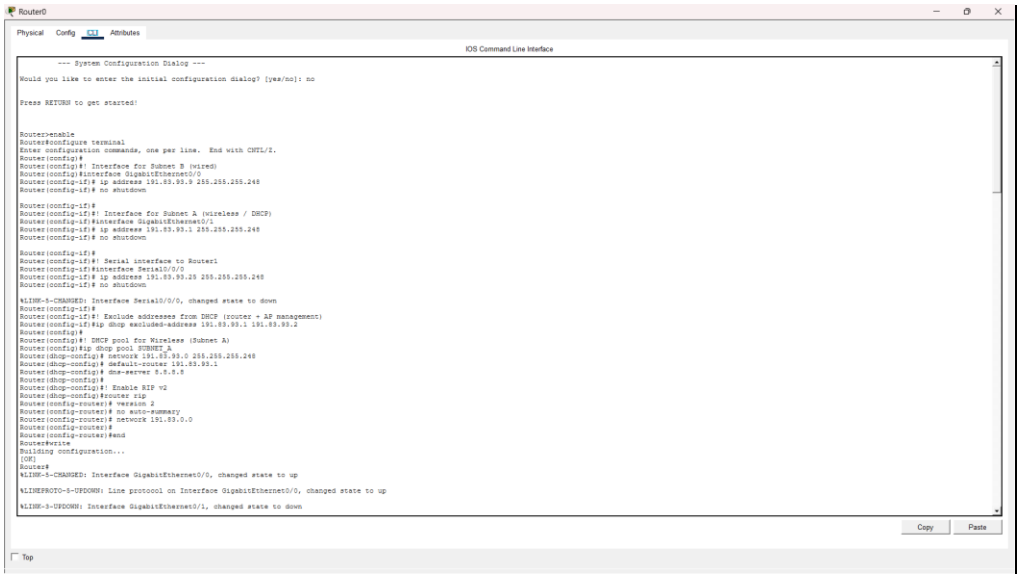

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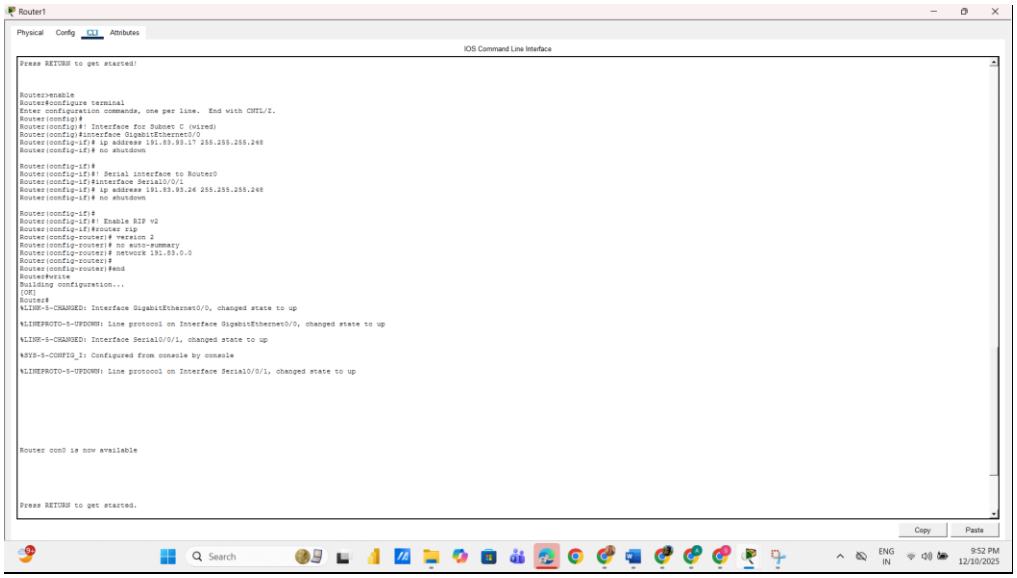

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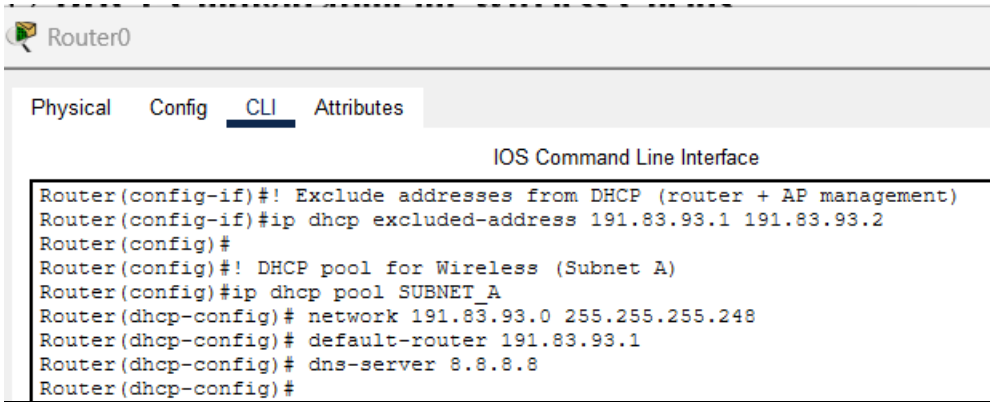

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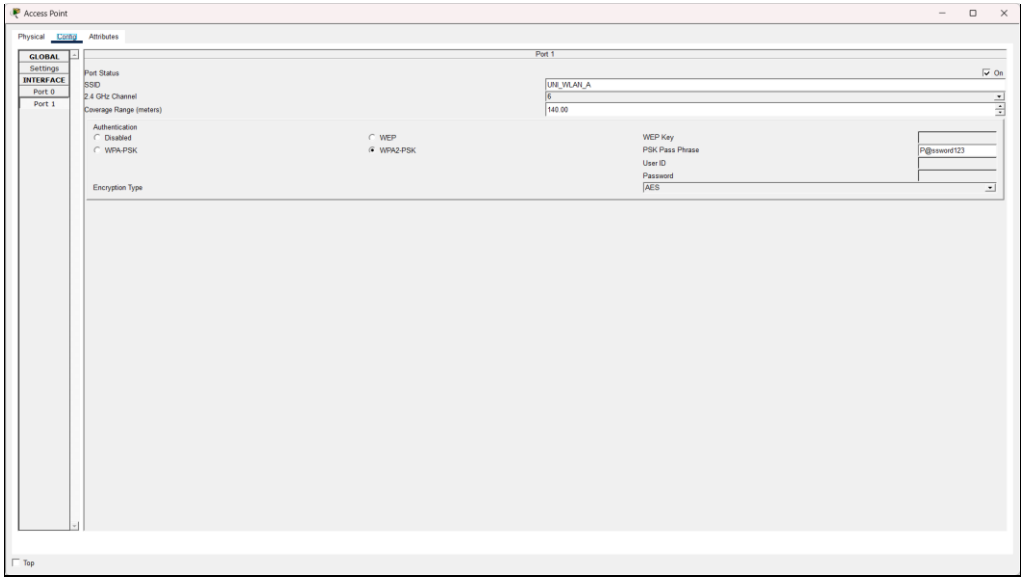

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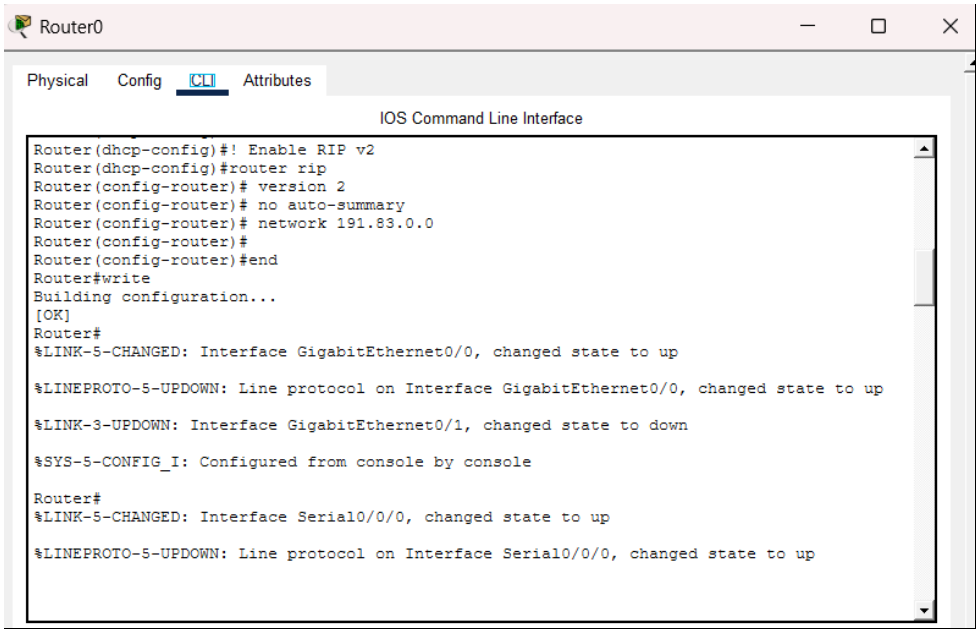

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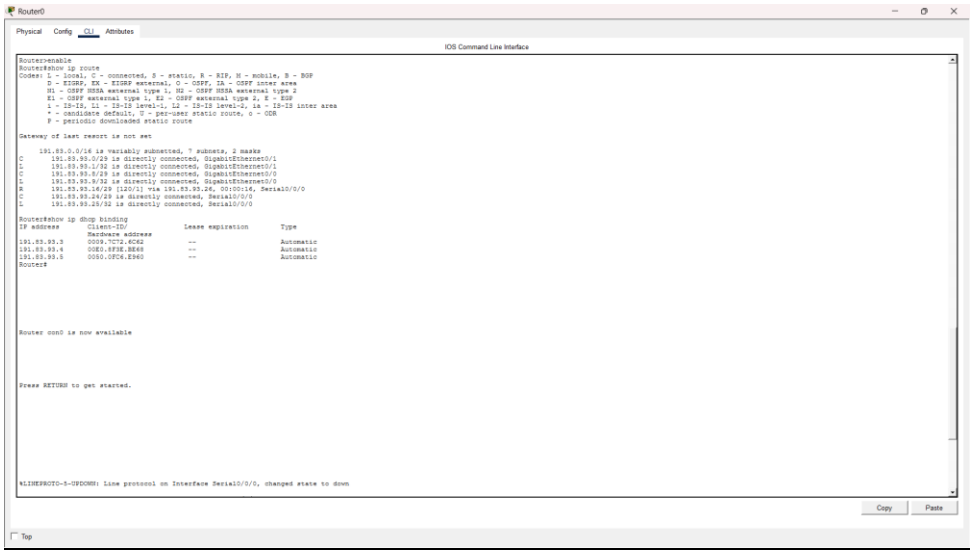

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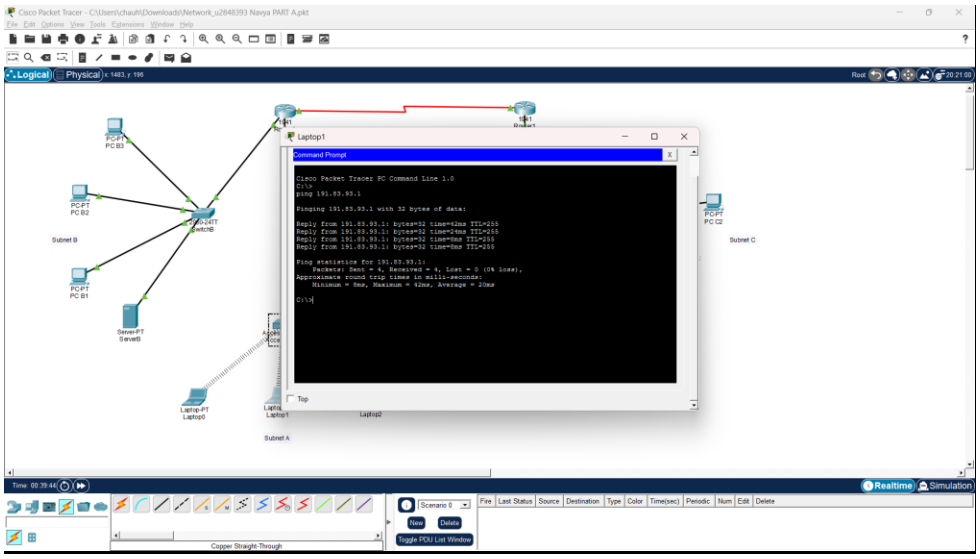

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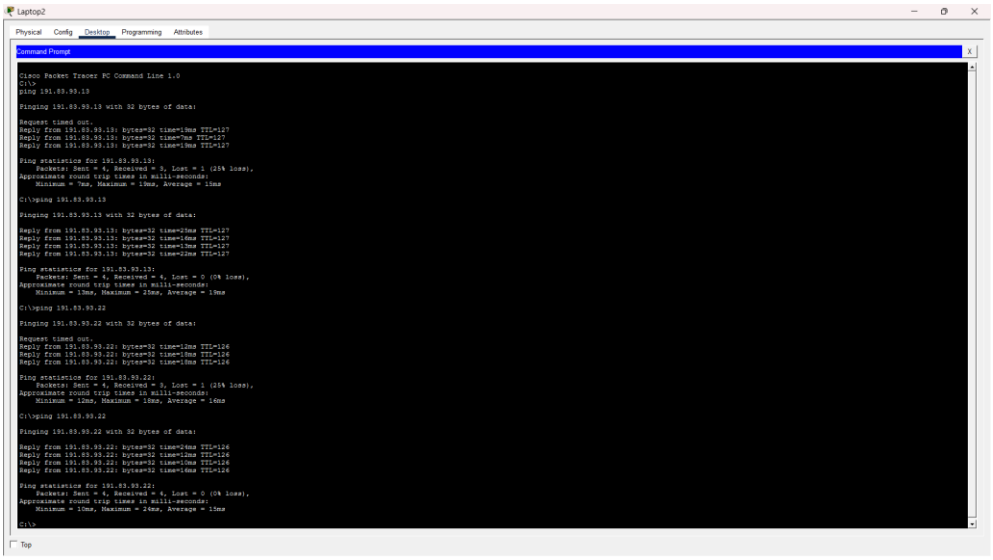

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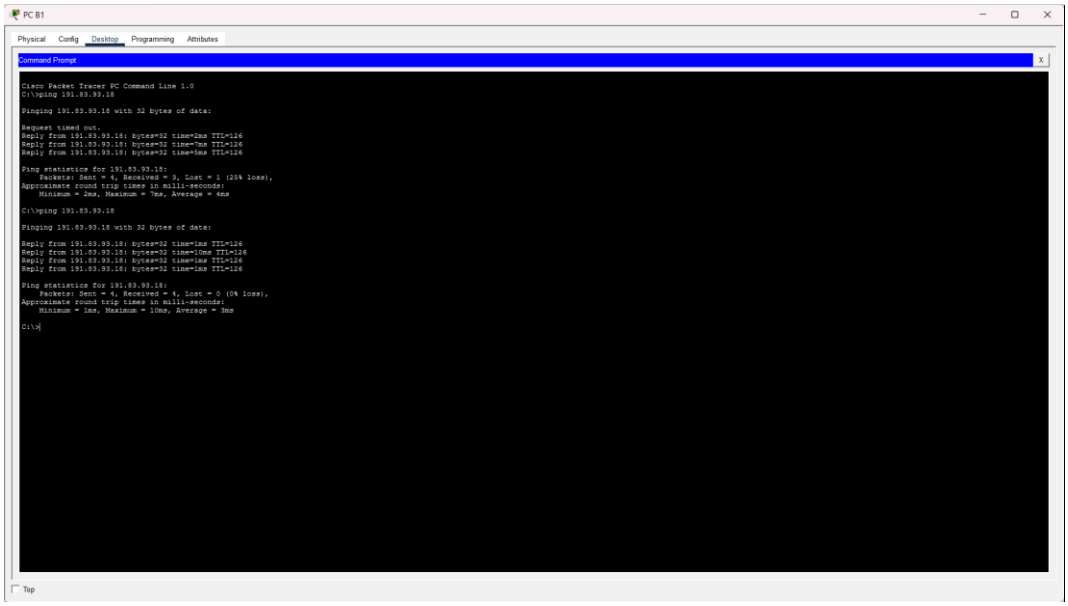

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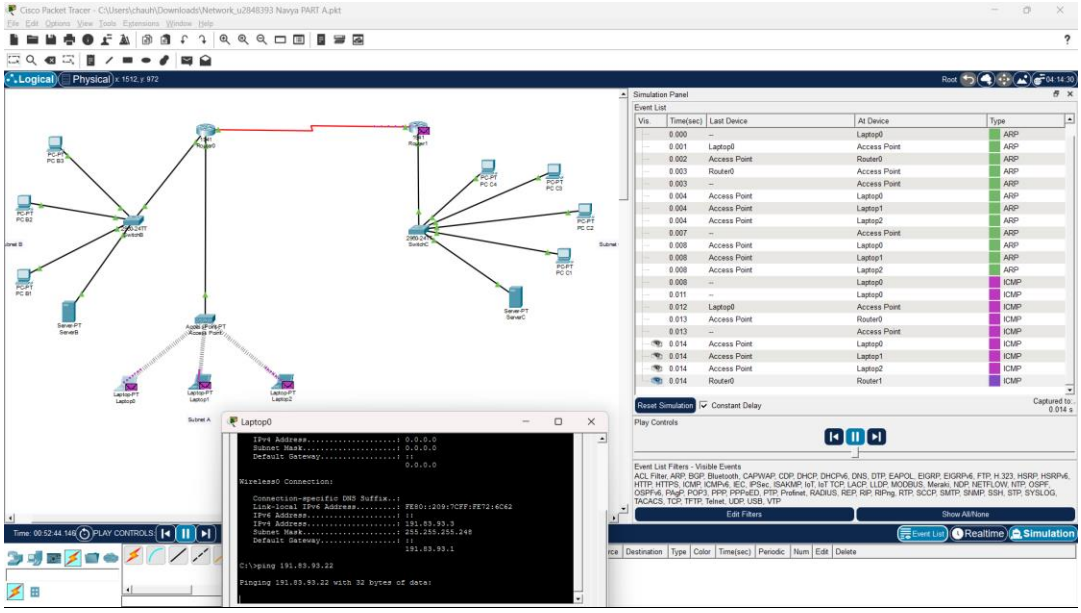

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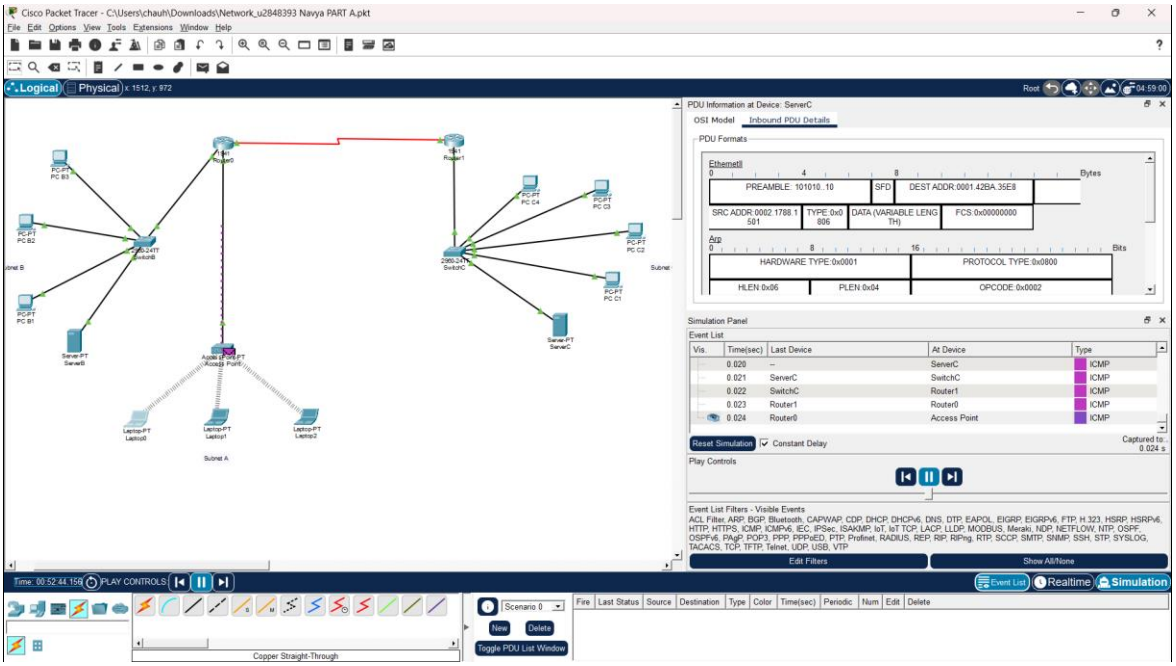

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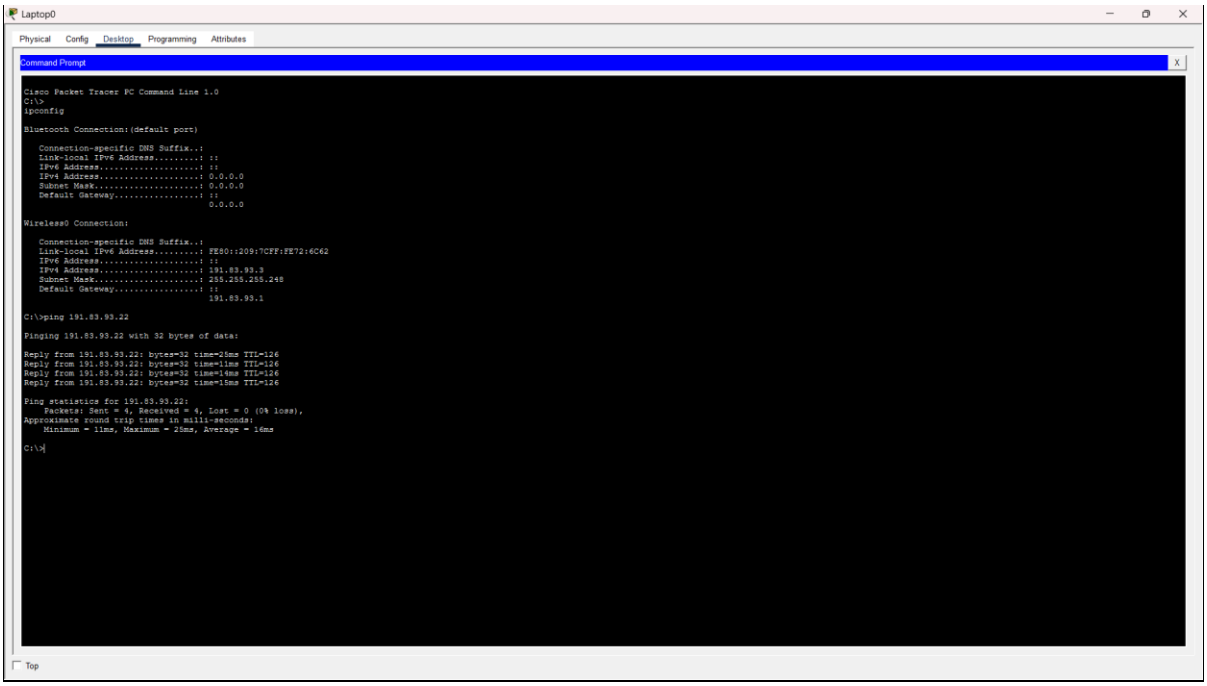

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